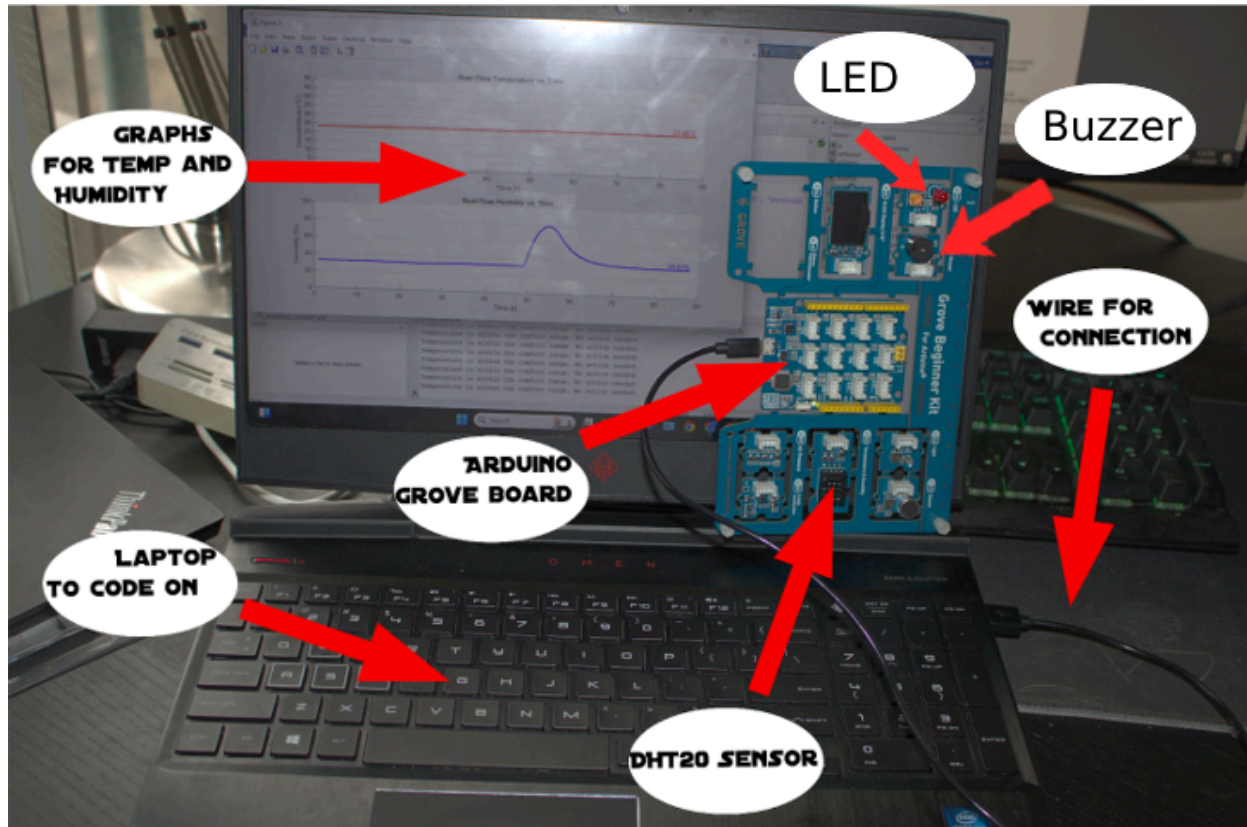


Bonus Project
Temperature Monitor Inside and Outside
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Intro

This project includes creating a real-time temperature monitoring system to reveal indoor and outdoor conditions. Using Arduino Grove board & DHT20 sensor, the system collects temperature and humidity data, displaying it on graphs. It provides live updates, making it ideal for tracking environmental changes. The project creates practical applications in energy efficiency and climate control while introducing concepts in sensor integration and data visualization.

CONTEXT

Use the Computing Environment:

This project uses MATLAB to code and show a temperature and humidity monitoring system, applying principles from science, math, and engineering. The system collects real-time environmental data using the DHT20 sensor and processes it in MATLAB to display temperature and humidity plots. Scientific data, mathematical modeling, and engineering systems are combined to create a practical simulation of real-world environmental monitoring.

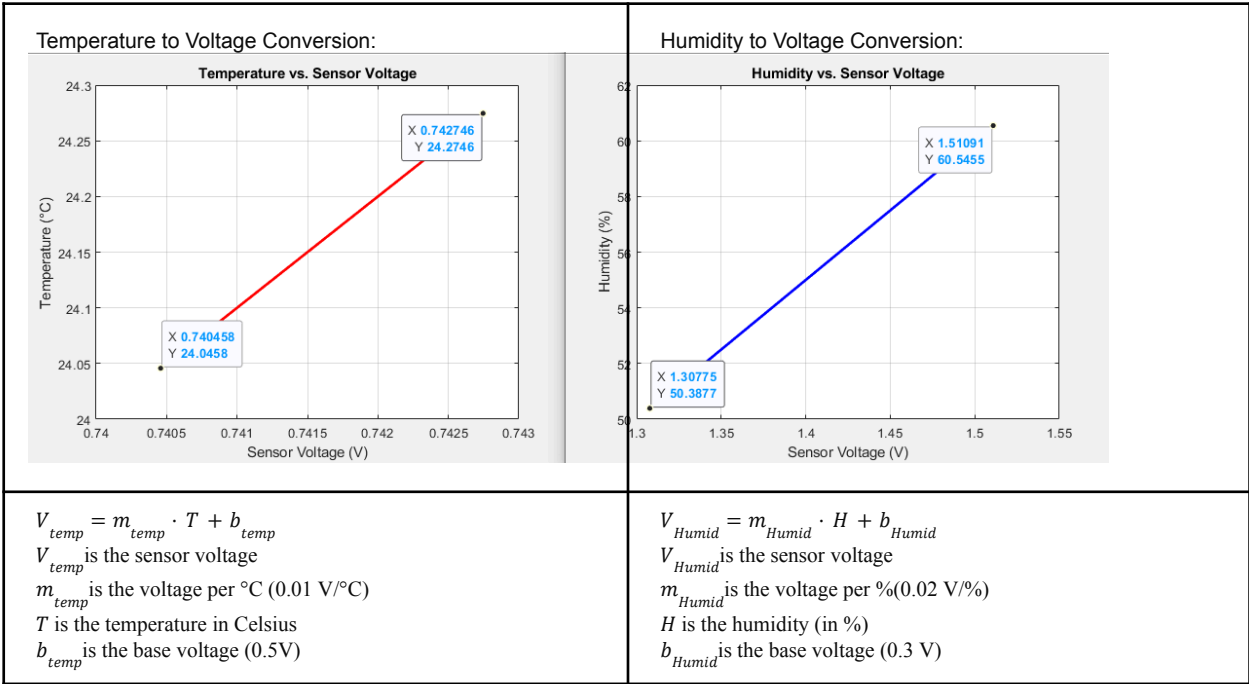
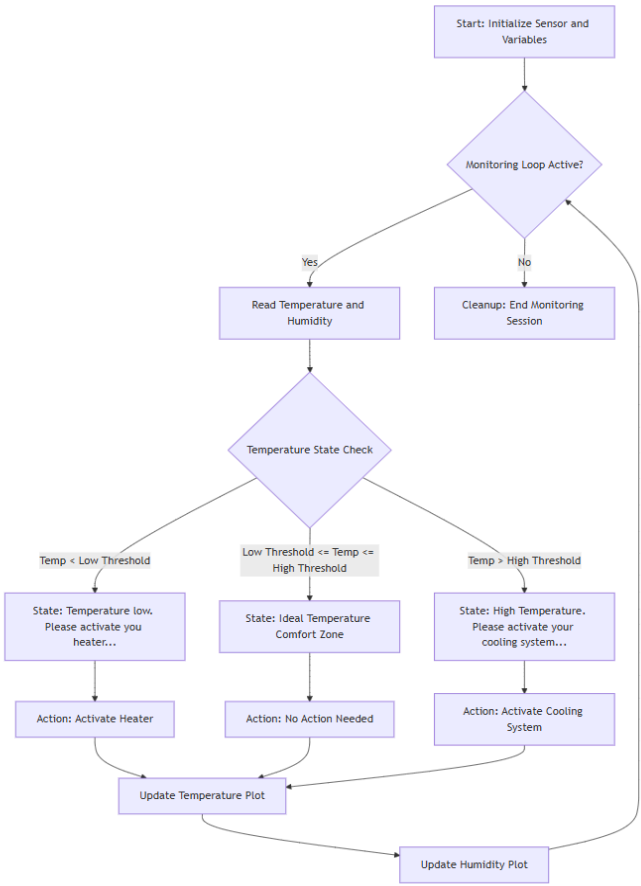
What and Why: SDG Topic

What: This project feels the most addressed to SDG 7: Affordable and Clean Energy because It is a tool to monitor indoor and outdoor temperatures and humidity. The data can show when it needs heating, cooling or when it is in a neutral warmth state and reducing energy consumption..

Why: Energy-efficient climate control systems are crucial for reducing dependency on non-renewable energy sources, lowering carbon emissions, and making clean energy more accessible and affordable. By implementing this project, the person reading this gains awareness of how data-driven graphs contribute to sustainable energy management and environmental protection.

TECHNICAL REQUIREMENTS

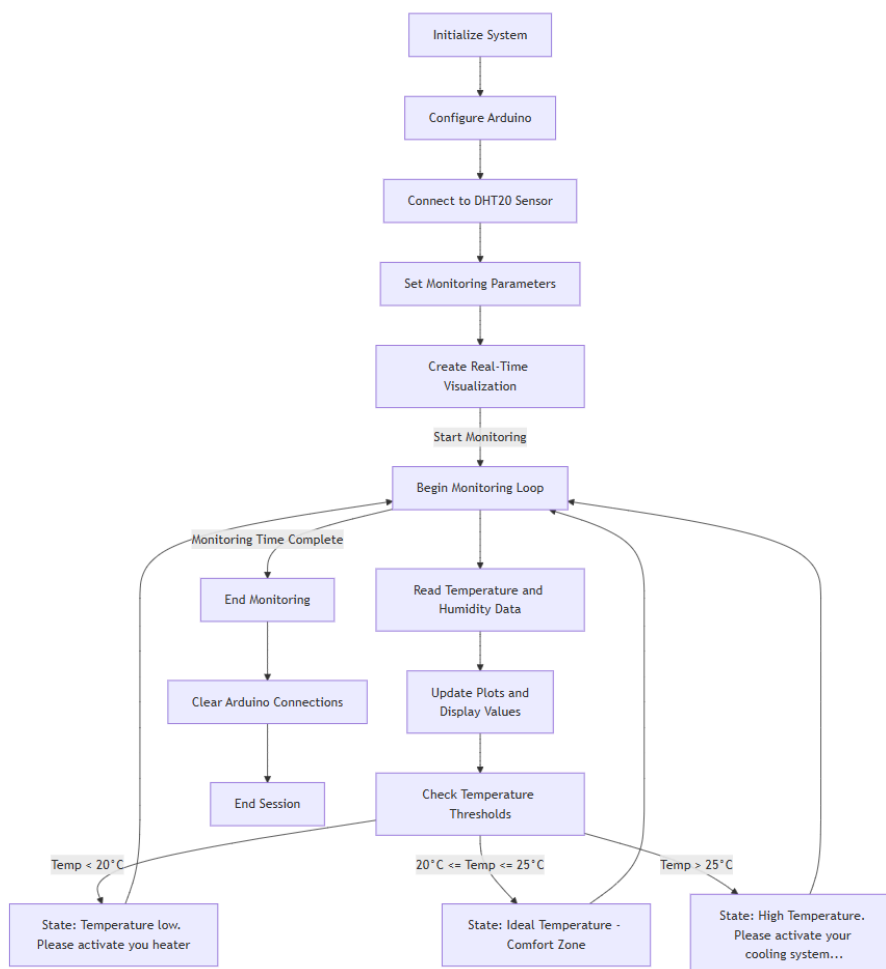
The list of things that the system should do is it has real-time monitoring where it collects and displays temperature and humidity in real-time. It has visualization where it plots the temperature and humidity on separate graphs concerning time on the x-axis. The system uses sensor integration, where it uses a file from Eric Prandovsky for the DHT20 sensor for temperature and humidity readings, using MATLAB with an Arduino Grove Nano board interface. The data on the graph updates every 2 seconds to match the sampling rate of the DHT20 sensor. The system's temperature graph specifications are set to 0°C and 50°C, and the humidity ranges between 0% to 100%. The Arduino communicates with MATLAB over I2C. The mathematical equations that are relevant here are in the dht20.m file where (relative humidity = humiditybits/10485.76), the humidity bits is the 20-bit raw humidity value gained from the sensor, and the 10485.76 factors normalize the raw value to get the humidity as a percentage. The second equation is also in that dht20.m file where its (temperature = (((temperatureBits/2^20)*200)-50)). The temperature bit is the 20-bit raw temperature value from the sensor, and the 2^20 scales the raw temperature data to a 20-bit range. The multiplication (200) and subtraction of 50 converts it to Celsius.



The system shows how the data continuously reads from the DHT20 sensor, which shows the change in voltages for both temperature and humidity. The Temperature vs. Voltage shows how sensor voltage changes with temperature. Humidity vs. Voltage shows how sensor voltage changes with humidity, which allows us to visualize the mathematical relationship between the sensor's output on Voltage and the measurements. This is helpful if we use this in calibration or further analysis thinking from an engineering perspective.

Procedure:

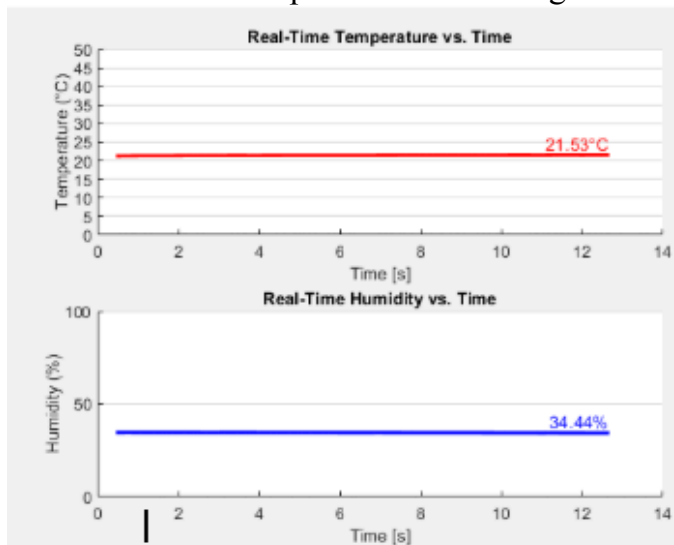
I first began making my initialization for the hardware and libraries I have used in the script, which was first writing my connection to the Arduino board, which is `a = arduino('COM3', 'Nano3', 'Libraries', {'I2C'})`, which allows me to communicate with the sensors. Next, I connected with the DHT20 sensor that reads the temperature and humidity using the DHT20 library. I have also added a figure with two subplots that display real-time temperature and humidity data. I made my real-time data monitoring by using animated lines to plot temperature and humidity in separate subplots: `hTemp` and `hHumid`. This loop continuously reads the data from the sensor, updating the plots with the latest values in gives. I made a state machine showing a temperature monitor. I predefined it with three key states: Low temperature. If the temperature falls below the low threshold, it asks to activate the heater. Ideal Temperature: No action is needed if the temperature is within the comfort zone. Last, High Temperature: If the temperature exceeds the threshold, it asks you to turn on your cooling system. Based on the sensor readings, if you're outside, the temperature drops drastically, but inside, it remains at an ideal temperature. Then, I wrote the data display in which the temperature and humidity values are continuously updated in the graphs with the temperature being in Celsius and humidity in %. I added text handles to update every loop iteration by deleting the old values and adding new ones to the graph for temperature and humidity. Last of all, at the end of the code, After completing the monitoring session, the code cleans up by clearing the Arduino connection and displaying a completion message. The libraries I used were I2C, the communication between the Arduino and the sensors, and dht20, which is for interfacing with the DHT20 sensor to read temperature and humidity data.



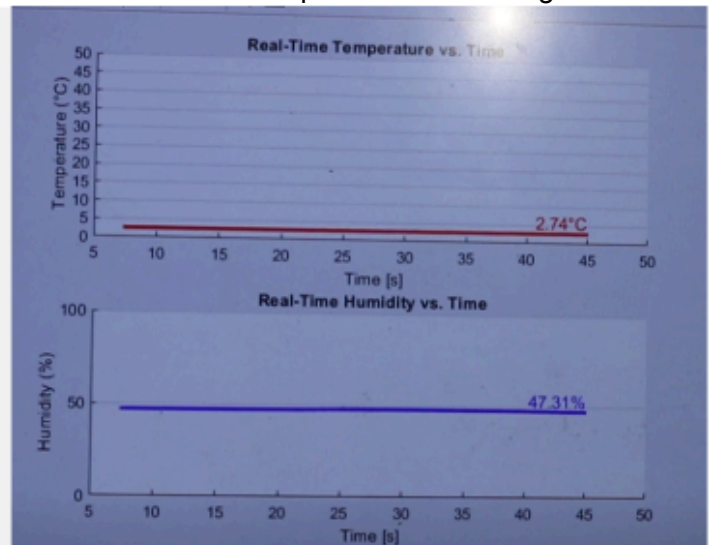
-----Flowchart-----

This project applies fundamental programming constructs to manage real-time data from a DHT20 sensor. Variables and Expressions: Variables like a temp, and humid store sensor data. Expressions such as `elapsedTime = toc(startTime)` calculate the time elapsed for monitoring. Control Structures: Conditionals: These are used to check the temperature and set the system state (e.g., heating is needed, cooling is required, or comfort zone). Loops: A while loop runs the monitoring process, continuously checking data and updating the visualization until the specified time is reached.

Indoor temperature monitoring



Outdoor temperature monitoring



Temperature is within the comfort range. No action needed. Temperature is low. Please activate you heater.

I tested if the system worked by leaving it indoors and inside, as shown in the graph. It shows the temperature and humidity. The left side of one of the graphs shows indoor monitoring, which shows that the temperature is within the comfort range, the moisture is one-third, the outdoor temperature is almost in the negative range, and the humidity is virtually half. While writing the system for monitoring temperature and humidity, I ran into some issues writing the code; the first was finding the sensor file to initialize the DHT20 sensor, which I searched online and found the code made by Eric Prandovsky, which then later found a video made by our Professor showing how the sensor works and implementing into the code. I had another error debugging the code, where the value on the graph, the blue and the red values, kept showing after each iteration, ruining the graph without showing the real-time graph. I figured it out by asking chat GPT what was wrong with my code and advising me. It is recommended that empty arrays be used to manage text handles, that the current value be added dynamically, and that the previous text object be deleted to maintain a clean graph. (if ~isempty(tempTextHandles) delete(tempTextHandles); % Delete previous text objects end tempTextHandles = text(elapsedTime, temp, sprintf('%.2f°C', temp), ... 'Color', 'r', 'VerticalAlignment', 'bottom', 'HorizontalAlignment', 'right');). I implemented the following code to address the issue for temperature values, and the TextHandles creates a text label at the coordinates (elapsed time, temp) on a plot. The label displays the temperature value temp formatted as %.2f°C with the text colour red ('Color', 'r') and aligns it vertically at the bottom and horizontally to the right of the specified point(The same goes for the humid textHandle). ChatGPT's advice provided the conceptual framework for this solution. I adapted and tested the code to ensure it worked for both temperature and humidity plots.

CONTINGENCY

When I was finding information on what I wanted to code for my project, I found it very interesting to code my temperature sensor onto my Oled display using Aradhya Chawl to initialize the Oled and use Eric's Prandovsky together to show the temperature on the Oled display. Still, I encountered some errors and time constraints where my code would show my temperature but not my humidity, and then it would crash my whole MATLAB, which I had to reset, so I had a time constraint. I needed to change my thinking and think of another solution. While the final result was functional, the original plan required a lot of thinking priorities and adjusting my workflow. What I would do differently is I would have conducted a more thorough practical approach to identify potential errors that would have occurred. Looking ahead to ENG 4000, I plan to bring more attention to my future detailed projected planning, including backup strategies if something doesn't work for critical components. Strengthen my ability to stay aligned with the main objects in the project. Last of all, I would use the lessons learned in debugging and problem-solving to bring new challenges with more confidence to be able to solve problems.

ADDITIONAL MATERIAL

This bonus project surpasses its expectations by addressing critical societal, environmental, and technological challenges through an innovative and combinative approach. The societal impact of temperature and humidity monitoring systems shows their usefulness in supporting farmers in making well-informed decisions, residential energy use, and water conservation. It shows how affordable technology can directly improve the quality of life, especially in rural areas; this shows environmental responsibility because it encourages sustainable practices by enhancing efficiency. It aligns with what I chose for SDG 7: Affordable and Clean Energy by showing the potential for renewable energy-powered systems to reduce their environmental impact and promote sustainability in everyday applications. In an online environment and coding, we display real-time data graphs; the projects use advanced data visualization techniques with scientific and engineering techniques, showing the real-world applications and that this project can perform well under increasing demands and future implementation potential in broader energy environmental monitoring systems. From a holistic perspective, this project lies in a social, political, and environmental perspective by revealing how renewable energy solutions and affordable technology can address global challenges like energy poverty and climate change. This project shows innovative design, real-world applications, and global sustainability where it exceeds its expectations beyond technical requirements to make a meaningful impact.

Conclusion

This bonus project shows a combination of real-time temperature and humidity monitor data with data visualization, showing technical and practical challenges and providing actionable insights for energy-efficient environmental management. It aligns with the topic SDG 7, which shows us scalable, low-cost solutions for sustainable living. Combining temperature/humidity to SDG 7, this innovative design as an undergraduate engineer with environmental awareness, the project provides global challenges and potentially exceeds expectations.

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